

Isolation of Fungi from Textile Effluent and Production of Bio-compost for Seed Germination

A MINI PROJECT REPORT



Submitted by

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in partial fulfilment for the award of the degree of

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In

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APRIL 2027**

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Rethika V M



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PO7: Environment and Sustainability

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ABSTRACT

The textile industry is one of the major contributors to industrial water pollution due to the discharge of untreated effluents containing synthetic dyes, organic pollutants, and toxic chemicals. These contaminants adversely affect aquatic ecosystems, soil quality, and agricultural productivity. The present study focuses on the isolation of fungal species from textile effluent and the evaluation of their potential in the biodegradation of textile pollutants and enhancement of seed germination.

Textile effluent samples were collected and subjected to microbial isolation using standard microbiological techniques. The isolated fungal strains were cultured, identified, and utilized for the biological treatment of textile effluent under controlled laboratory conditions. The treated effluent was subsequently evaluated for its influence on the germination and early growth of green gram (*Vigna radiata*) seeds at different concentrations. The study also included protein profiling using SDS-PAGE to analyze the biochemical characteristics of the isolated fungal cultures.

The experimental results demonstrated that fungal treatment effectively reduced the toxicity of textile effluent and improved seed germination and plant growth compared to untreated samples. Higher concentrations of biologically treated effluent exhibited better germination performance, indicating the effectiveness of fungal biodegradation. These findings highlight the potential application of indigenous fungal isolates as an eco-friendly, sustainable, and cost-effective approach for textile wastewater remediation while supporting agricultural productivity.

This study emphasizes the significance of fungal bioremediation as an environmentally sustainable alternative for industrial wastewater management and promotes the reuse of treated textile effluent for agricultural applications.

Keywords:

Textile effluent, Fungal isolation, Bioremediation, *Aspergillus spp.*, Microbial degradation, Seed germination,

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LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Full Form
BOD	Biological Oxygen Demand
COD	Chemical Oxygen Demand
PDA	Potato Dextrose Agar
MEA	Malt Extract Agar
MEB	Malt Extract Broth
SDS	Sodium Dodecyl Sulfate
SDS-PAGE	Sodium Dodecyl Sulfate Polyacrylamide Gel Electrophoresis
DNA	Deoxyribonucleic Acid
PCR	Polymerase Chain Reaction
HPLC	High Performance Liquid Chromatography
GC-MS	Gas Chromatography–Mass Spectrometry
TDS	Total Dissolved Solids
TSS	Total Suspended Solids
EC	Electrical Conductivity
pH	Potential of Hydrogen
rpm	Revolutions Per Minute
°C	Degree Celsius
CFU	Colony Forming Unit
LCB	Lactophenol Cotton Blue
ETP	Effluent Treatment Plant
UV	Ultraviolet
mg/L	Milligrams per Liter
mL	Milliliter
g/L	Grams per Liter

1. INTRODUCTION

1.1 Textile Industry and Environmental Pollution

The textile industry is one of the largest manufacturing sectors in the world and plays a significant role in economic development by producing fabrics, garments, and other textile products. Textile processing operations such as desizing, scouring, bleaching, dyeing, printing, and finishing require large quantities of water and chemicals, resulting in the generation of significant volumes of wastewater [1].

Textile effluents contain dyes, heavy metals, phenols, surfactants, salts, and other toxic organic compounds that are difficult to degrade. When discharged without proper treatment, these pollutants contaminate surface water, groundwater, and soil, causing severe environmental pollution and posing risks to aquatic organisms and human health [2], [3].

The colored wastewater released from textile industries reduces light penetration in water bodies, affecting photosynthesis and dissolved oxygen levels. In addition, toxic chemicals and heavy metals accumulate in living organisms through the food chain, leading to long-term ecological and health impacts [3], [4].

Conventional treatment methods for textile wastewater are often expensive and may generate secondary pollutants. Therefore, biological treatment using microorganisms, particularly fungi, has emerged as an effective and eco-friendly alternative. Fungi possess the ability to degrade complex dye molecules and reduce the toxicity of textile effluents, making them suitable for bioremediation applications [5], [6]



Figure 1.1 Textile Effluent sample

1.2 Characteristics of Textile Effluent

Textile effluent is a complex wastewater generated during various textile processing operations such as desizing, scouring, bleaching, dyeing, printing, and finishing. Its composition varies depending on the type of fabric, dyes, chemicals, and manufacturing processes employed. Generally, textile effluents are characterized by high color intensity, alkaline pH, elevated chemical oxygen demand (COD), biological oxygen demand (BOD), total dissolved solids (TDS), total suspended solids (TSS), and the presence of toxic organic and inorganic pollutants [1], [2].

The wastewater contains synthetic dyes, surfactants, detergents, salts, starch, oils, grease, phenolic compounds, heavy metals, and other auxiliary chemicals used during textile processing, many of which are resistant to biodegradation and remain in the environment for long periods [2], [3]. The intense color of textile effluent reduces light penetration into water bodies, thereby affecting photosynthesis and decreasing dissolved oxygen levels required for the survival of aquatic organisms [3], [4].

High BOD and COD values indicate the presence of large amounts of biodegradable and non-biodegradable organic matter, leading to oxygen depletion and deterioration of water quality [4], [5]. The high concentration of dissolved salts and suspended solids increases the salinity and turbidity of water, while the alkaline nature of the effluent, resulting from the use of sodium hydroxide and other processing chemicals, adversely affects aquatic ecosystems and reduces the efficiency of biological treatment processes [5], [6].

Owing to these characteristics, textile effluent requires effective treatment before discharge into the environment, and biological methods using fungi and other microorganisms have emerged as sustainable and eco-friendly alternatives for pollutant removal remediation[6],[7]

1.3 Impact of Textile Effluent on Environment

The discharge of untreated textile effluent into the environment causes serious ecological and public health problems [1]. Textile wastewater contains synthetic dyes, heavy metals, salts, and toxic organic compounds that contaminate surface water, groundwater, and soil [2]. The intense color of the effluent reduces light penetration into water bodies, affecting photosynthesis and decreasing dissolved oxygen levels essential for aquatic life [3]. High biological oxygen demand (BOD) and chemical oxygen demand (COD) further deplete dissolved oxygen, leading to the death of fish and other aquatic organisms [4]. Heavy metals and toxic chemicals accumulate in plants and animals through the food chain, posing health risks to humans [5]. Continuous discharge of textile wastewater also reduces soil fertility, affects crop productivity, and alters the physicochemical properties of water and soil [6]. Some dyes and chemicals present in textile effluents are toxic, mutagenic, and potentially carcinogenic, making effective treatment essential before disposal [7]. Therefore, the proper management and treatment of textile effluents are necessary to protect environmental quality and ensure sustainable industrial development [8].

1.4 Fungi in Bioremediation

Synthetic Fungi play a significant role in the bioremediation of textile effluents because of their ability to degrade a wide range of organic pollutants and synthetic dyes [1]. They produce extracellular enzymes such as laccase, lignin peroxidase, and manganese peroxidase, which break down complex dye molecules into simpler and less toxic compounds [2], [3]. Fungal species such as *Aspergillus*, *Penicillium*, *Trichoderma*, and *Phanerochaete chrysosporium* are widely used in the treatment of textile wastewater due to their high biodegradation efficiency [3], [4]. These fungi can remove color, reduce biological oxygen demand (BOD) and chemical oxygen demand (COD), and detoxify harmful pollutants present in textile effluents [4], [5]. Unlike conventional treatment methods, fungal bioremediation is cost-effective, eco-friendly, and produces minimal secondary pollution [5], [6]. In addition, fungal biomass can be utilized for compost production, contributing to sustainable waste management and improving soil fertility [6]. Therefore, the use of fungi in bioremediation is considered a promising approach for the efficient treatment of textile wastewater and environmental protection [7].

1.5 Mechanism of Fungal Degradation

Fungal degradation is an effective biological process in which fungi utilize organic pollutants as a source of carbon and energy, thereby reducing their toxicity [1]. Fungi secrete extracellular enzymes such as laccase, lignin peroxidase, manganese peroxidase, and cellulase that break down complex dye molecules and other pollutants into simpler compounds [2], [3]. Initially, pollutants are adsorbed onto the fungal cell wall through a process known as biosorption [3]. The pollutants are then absorbed into the fungal cells, where enzymatic reactions convert them into less toxic intermediates [4]. These compounds undergo further biodegradation and biotransformation, resulting in the formation of simpler molecules such as carbon dioxide, water, and inorganic salts [5]. The extensive mycelial network of fungi provides a large surface area for efficient pollutant removal and degradation [6]. This process reduces the color, toxicity, biological oxygen demand (BOD), and chemical oxygen demand (COD) of textile wastewater, making it environmentally safe for disposal or reuse [6], [7]. Therefore, fungal degradation is considered a sustainable, eco-friendly, and cost-effective approach for the bioremediation [7],[8].

1.6 Seed Germination and Plant Growth Promotion

Typha Seed germination is the process by which a seed develops into a healthy seedling under favourable environmental conditions [31]. Healthy seed germination is essential for proper plant growth and higher crop productivity [31]. The application of fungal-treated compost or biologically treated textile effluent can improve soil quality by increasing the availability of nutrients required for plant growth [13, 32]. Beneficial fungi help in the decomposition of organic matter and release essential minerals into the soil, which support seed germination and root development [13]. In addition, fungal activity improves soil fertility and enhances water-holding capacity, creating a suitable environment for plant growth [13, 32]. Seeds grown in nutrient-rich soil generally show better germination percentage, increased root length, shoot length, and overall seedling vigour [31]. Therefore, the use of fungi in bioremediation not only helps reduce environmental pollution but also promotes sustainable agriculture by improving seed germination and healthy plant growth [13, 31, 32].

1.7 Problem statement

Textile industries release wastewater containing harmful dyes and chemicals that cause serious environmental pollution. Conventional treatment methods are expensive and may not completely remove these pollutants. Therefore, there is a need for an eco-friendly and cost-effective treatment method. This study focuses on isolating fungi from textile effluent and using them for bioremediation and seed germination to promote environmental sustainability.

1.8 Aim

The aim of this project is to isolate fungi from textile effluents to degrade pollutants and heavy metals, while also enhancing seed germination and soil fertility. This provides a sustainable bioremediation strategy and an eco-friendly alternative to conventional treatment methods.

1.9 Objectives

- To isolate fungi from textile effluent.
- To identify fungal strains capable of degrading pollutants present in textile effluent.
- To evaluate the role of isolated fungi in promoting seed germination.
- To utilize fungal strains for the bioremediation of contaminated textile wastewater

1.10 Scope of study

This study focuses on isolating fungi from textile effluents to evaluate their ability to degrade pollutants such as dyes, organic compounds, and heavy metals, while also enhancing seed germination and soil fertility. It integrates microbiology, environmental science, and agronomy to develop an eco-friendly bioremediation strategy that reduces pollution and supports sustainable agriculture.

1.11 Significance of the Study

This study is significant because it shows that fungi isolated from textile effluents can both degrade pollutants and promote seed germination, offering a dual benefit for bioremediation and sustainable agriculture. As the document notes, "Fungi possess diverse metabolic capabilities and plant growth-promoting traits that make them well-suited for addressing the environmental challenges associated with textile wastewater pollution."

2. REVIEW OF LITERATURE

2.1 Textile wastewater characteristics

Textile wastewater is generated during various processing operations and contains dyes, organic compounds, salts, and other pollutants that require effective treatment. A critical review on the production, characterization, and treatment of textile effluents was presented by Haley *et al.* (2014) [3], who highlighted the challenges associated with textile wastewater and discussed different treatment methods, including hydrogen peroxide with UV-C for efficient decolorization. The authors also emphasized the importance of understanding the composition of textile effluents and selecting suitable treatment technologies based on COD content to minimize environmental pollution.

The potential of *Bacillus cohnii* strain RKS9 in the decolorization and detoxification of Congo red dye and textile wastewater was demonstrated by Kishor *et al.* (2021) [29]. The study reported significant reductions in colour, COD, BOD, and phenol concentration, indicating the effectiveness of microbial treatment. The involvement of enzymes such as manganese peroxidase, azoreductase, lignin peroxidase, and laccase further confirmed its potential as an eco-friendly approach for textile wastewater remediation.

2.2 Environmental impact of textile effluent

The environmental impact of textile effluents on agricultural crops has been widely studied. Kaushik *et al.* (2005) [11] observed that increasing textile effluent concentration affected the growth performance of wheat cultivars, although properly diluted effluent could provide nutrients beneficial for irrigation. Their findings emphasized the need for careful management of textile wastewater before its use in agriculture.

The harmful effects of textile dyes on human health and the environment were also discussed by Hassan and El Nemr (2017) [2], who reported that untreated dye-containing effluents are major sources of water pollution. The study stressed the importance of adopting effective and sustainable treatment methods to reduce the environmental and health risks associated with textile industry wastewater.

2.3 Conventional treatment method

Several conventional treatment methods have been developed to remove dyes from textile wastewater. The application of electrocoagulation for the decolorization of textile dye solutions containing C.I. Basic Yellow 28 was investigated by Daneshvar *et al.* (2006) [4]. The study used artificial neural networks (ANN) to evaluate the effect of operational parameters on dye removal and concluded that electrocoagulation is an effective method for treating dye-containing wastewater.

Advanced oxidation is another effective treatment technique for textile wastewater. Safarzadeh *et al.* (2001) [23] reported that the H₂O₂/UV process produces hydroxyl radicals capable of degrading hazardous pollutants and achieving high COD removal without sludge formation. The study also demonstrated excellent decolorization efficiency for various textile dyes, indicating that this method is a promising option for wastewater treatment.

2.4 Fungal Bio-remediation of Textile Effluent

Fungal bioremediation has emerged as an eco-friendly and efficient approach for treating textile wastewater. The decolorization of textile dyes using *Aspergillus tamari*, mixed fungal culture, and *Penicillium purpurogenum* was demonstrated by Ramalingam *et al.* (2010) [1]. Their study showed that these fungal species effectively degraded textile dyes and significantly reduced the colour of textile effluents, highlighting their potential application in sustainable bioremediation.

2.5 *Aspergillus* species in wastewater treatment

Aspergillus has been widely investigated for its ability to remove textile dyes from wastewater. A study conducted by Salem *et al.* (2019) [28] isolated *Aspergillus niger* D2-1 from contaminated soil collected from textile industries and demonstrated its high efficiency in decolorizing Reactive Yellow (4GL) and Reactive Red (4BL) dyes. The fungal isolate was identified through morphological characteristics and ITS gene sequencing. The study also reported significant reductions in colour, COD, and TDS, highlighting the potential of *Aspergillus niger* for the biological treatment of textile wastewater.

The efficiency of different fungal species in dye removal has also been investigated. Sivakumar *et al.* (2014) [27] evaluated *Aspergillus niger* and *Aspergillus flavus* for textile

dye decolorization and found that both fungi exhibited high dye removal efficiency under optimized conditions. Their findings demonstrated the suitability of these fungal species for sustainable colour removal from textile effluents.

2.6 Seed germination studies using treated effluent

The effect of fungal treatment on textile wastewater and its application in environmental remediation has been widely studied. Krishnamoorthy *et al.* (2017) [30] reported that mixed fungal cultures effectively degraded azo dyes into simpler compounds, achieving up to 95% dye removal. The study highlighted the potential of fungal treatment for improving textile wastewater quality and promoting environmentally sustainable bioremediation.

The influence of treated textile effluent on seed germination and plant growth was evaluated by Rehman *et al.* (2008) [31], who observed that treated effluent supported the early growth of winter vegetable crops. The study also emphasized that proper treatment of textile wastewater before agricultural use helps reduce its harmful effects while promoting sustainable crop production.

2.7 Research gap

Most previous studies have focused on the physicochemical characteristics of textile wastewater, dye decolorization, and pollutant removal using physical, chemical, and biological treatment methods. Several researchers have reported the effectiveness of fungal species such as *Aspergillus*, *Penicillium*, *Trametes*, and *Phanerochaete* in textile wastewater treatment. However, only a few studies have combined fungal isolation from textile effluent, biodegradation of textile pollutants, and evaluation of treated effluent on seed germination in a single investigation. Therefore, the present study focuses on isolating *Aspergillus* species from textile effluent, assessing their biodegradation potential, and evaluating the effect of fungal-treated textile effluent on the seed germination of green gram (*Vigna radiata*).

2.8 Conceptual framework of the proposed study

The conceptual framework of this study is based on the application of fungal bioremediation for textile wastewater treatment and its subsequent use in agriculture. Textile effluent is collected from textile industries, followed by the isolation and identification of *Aspergillus species*. The selected fungal isolate is cultured and used for the biodegradation of textile wastewater under different effluent concentrations. After treatment, the effluent is purified, analyzed for its quality, and applied to green gram (*Vigna radiata*) seeds to evaluate seed germination and plant growth. This framework demonstrates that fungal treatment can reduce pollutant load in textile wastewater while improving its suitability for agricultural reuse and sustainable environmental management.

3 MATERIALS AND METHODS

3.1 Study area and sample collection

This mini project study was conducted using textile wastewater collected from Agraharam, Erode, Tamil Nadu, a major textile processing area. The wastewater sample was collected in sterile containers and transported to the laboratory for analysis. The collected effluent was used for the isolation of *Aspergillus species*, biodegradation studies, and evaluation of the effect of fungal-treated effluent on the seed germination of green gram (*Vigna radiata*).

3.2 Microbial count of textile effluent

a) Serial Dilution

Mix the textile effluent and pipette 1 ml into a tube containing 9 ml of buffered peptone water. Mix carefully from this first dilution. Transfer 1 ml into the second dilution tube containing 9 ml of distilled water, and continue serial dilutions up to 10^{-1} , 10^{-2} , 10^{-3} , 10^{-4} , and 10^{-5} .

b) Plating

- Pour Plating

From each dilution, 1 ml of the sample was transferred to sterile Petri plates. Sterilized Potato Dextrose Agar (PDA) was poured onto the Petri plate and mixed thorough

- Spread Plating

Sterilized medium was poured onto the Petri plate and allowed to solidify. From each dilution, 0.1 ml of the sample was spread onto the surface of the medium. Using a sterilized L-shaped glass rod, the inoculum was spread evenly over the plate.

c) Incubation

Potato Dextrose Agar plates were incubated at 37°C for 24 hours, and then maintained at room temperature for 3–4 days.

d) Observation

After incubation, the growth of microorganisms was observed on the plates.

3.3 Isolation and identification of fungi

- Textile effluent samples were serially diluted, and the pre-serially diluted sample was used for inoculation.
- Malt Extract Agar (MEA) plates were prepared under sterile conditions.
- A sterile L-shaped bent glass rod was dipped in 95% ethyl alcohol and flame sterilized. The rod was allowed to cool for 10–15 seconds.
- The diluted sample was transferred onto the surface of the Malt Extract Agar plate.
- The inoculum was spread evenly over the agar surface using the sterile bent glass rod while rotating the Petri plate on a turntable.
- The inoculated plates were incubated in an inverted position at 25°C for 48–72 hours for fungal growth.
- After incubation, distinct fungal colonies were selected for identification.
- For identification, a drop of Lactophenol Cotton Blue (LPCB) stain was placed on a clean glass slide.
- A small portion of the fungal colony along with supporting agar was transferred onto the stain and teased gently into small fragments.
- A coverslip was placed over the preparation, and the slide was allowed to stand for about 50 minutes to absorb the stain.
- The stained fungal preparation was examined under a microscope to observe mycelial morphology and identify the fungal isolate.

3.4 Preparation of culture media

a.Malt extract agar

provides all the nutrients required by the microorganisms. This media is used to growth pure culture of *Aspergillus* species.

S.NO	Ingredients	g/l
1	Maltose	12.5
2	Dextrin	2.5
3	Glycerol	1.0
4	Peptocomplex	2.6
5	Agar	17

Table 3.1 Composite of Malt Extract Agar

To prepare 1 liter of Malt Extract Agar:

- Weigh the required amounts of Maltose (12.5 g), Dextrin (2.5 g), Glycerol (1.0 g), Peptocomplex (2.6 g), and Agar (17 g).
- Dissolve all the ingredients in distilled water and make up the volume to 1 liter.
- Heat the mixture until the agar dissolves completely.
- Sterilize the medium by autoclaving.
- Pour the sterilized medium into sterile Petri plates and allow it to solidify.
- The prepared Malt Extract Agar plates are then used for the isolation and maintenance of pure cultures of *Aspergillus species*.

b.Malt extract broth

This is a liquid media generally used to culture microorganisms. It provides all the basic nutrients required by the microorganisms. This contains Malt Extract, Maltose, Glucose and Yeast extract. The inoculated medium incubated at 37°C for 48–91 hours on shaker.

S.NO	Ingredients	g/l
1	Malt Extract	0.9
2	Maltose	0.27
3	Glucose	0.9
4	Yeast extract	0.8
5	pH	4.7

Tabel 3.2 Composition of Malt Extract broth

To prepare 1 liter of Malt Extract Broth:

- Weigh the required ingredients:
- Malt Extract – 0.9 g
- Maltose – 0.27 g
- Glucose – 0.9 g
- Yeast Extract – 0.18 g
- Dissolve all the ingredients in distilled water and make up the volume to 1 liter.
- Adjust the pH of the medium to 4.7.
- Dispense the prepared broth into suitable culture flasks (e.g., 125 mL Erlenmeyer flasks).
- Sterilize the medium by autoclaving under standard conditions.
- After cooling, inoculate the broth with the fungal culture (*Aspergillus* species).
- Incubate the inoculated Malt Extract Broth at 37°C for 48–91 hours on a shaker to promote fungal growth.
- The prepared broth culture can then be used for maintenance, sporulation, and further experimental studies involving *Aspergillus* species

3.6 Inoculum development

Inoculate large quantities of degradation medium are needed in high numbers, since a loopful of culture cannot meet the requirements for degradation. Therefore, microbes from the stock culture are transferred into a larger volume of medium for inoculum development. The inoculated medium is then incubated under shake culture conditions at 280 rpm for 3–7 days. After incubation, the developed inoculum is used for the degradation process.

3.7 Treatment of textile effluent

- The textile effluent was treated with fungal broth culture of *Aspergillus* species in different concentrations like 20%, 40%, 60%, 80%, and 100%.
- In this treatment, the concentration of effluent will vary, but the microbial concentration is constant.
- Microbial inoculated textile effluent was incubated at 27°C for 10–40 days on a shaker at 220 rpm

3.8 Purification of treated effluent

- Treated effluent will also have some waste debris.
- It was centrifuged at 5000 rpm for 10 minutes, and the supernatant was collected and diluted with water at various concentrations.

3.5 Seed germination

- The green gram seeds were surface sterilized in a 1% hypochloride solution and washed three times in distilled water.
- Then, the seeds were soaked with four different concentrations of textile effluent (20%, 40%, 60%, 80%, 100% and control).
- Seeds were inoculated in pots, and the respective dilution of effluent was added to maintain the moisture content of the soil.
- After 5 to 7 days, the seed germination percentage was calculated and compared with the control.

3.10 SDS-PAGE Procedure

The SDS-PAGE procedure was carried out by first cleaning and assembling the glass plates with spacer bars to prevent leakage. A 12% separating gel was prepared using distilled water, 30% acrylamide, 1.5 M Tris-HCl (pH 8.8), 10% SDS, 10% APS, and TEMED, poured between the glass plates, overlaid with distilled water, and allowed to polymerize for 30–60 minutes. After removing the water layer, a 5% stacking gel containing distilled water, 30% acrylamide, 0.5 M Tris-HCl (pH 6.8), 10% SDS, 10% APS, and TEMED was prepared, poured over the separating gel, and fitted with a comb to form sample wells. The protein samples were mixed with SDS sample buffer and heated at 100°C for 3 minutes before loading into the wells along with a molecular weight marker. The gel assembly was placed in an electrophoresis tank containing running buffer and run at 100–150 V until the tracking dye reached the bottom of the gel. Following electrophoresis, the gel was removed from the glass plates and stained with Coomassie Brilliant Blue R-250 for 2–3 hours or overnight. Finally, the gel was destained until clear protein bands became visible, and the separated bands were observed and documented for analysis.

3.11 Limitations

As a mini project, the study may be limited by short duration, small reactor size, restricted analytical facilities, and the use of simulated rather than real industrial effluent. These limitations should be explicitly acknowledged while discussing the finding

4.RESULTS AND DISCUSSION FRAMEWORK

4.1 Isolation of fungal colonies

Textile wastewater collected from Agraharam, Erode, was serially diluted and inoculated on Potato Dextrose Agar (PDA) medium. After incubation, distinct fungal colonies were observed. The colonies were sub-cultured repeatedly to obtain pure fungal isolates. The isolated fungi showed rapid growth with characteristic colony morphology suitable for further identification.



Figure 4.1 Isolation of fungal colonies

4.2 Identification of Fungal Isolates

The purified fungal isolate was identified as *Aspergillus spp.* based on colony morphology and microscopic examination using Lactophenol Cotton Blue (LPCB) staining. Microscopically, the isolate showed septate hyphae, conidiophores, vesicles, phialides, and chains of conidia, confirming its identity as *Aspergillus* species.

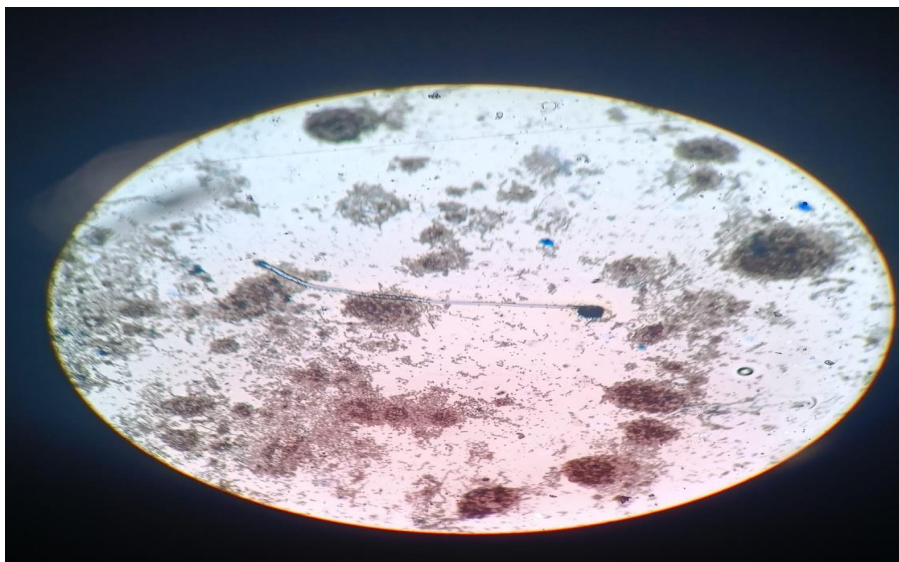


Figure 4.2 Identification of fungi

4.3 Effects of Fungal Treatment on Textile Effluent

The isolated *Aspergillus* species was inoculated into textile wastewater and incubated under suitable conditions. After treatment, a noticeable reduction in the colour intensity of the textile effluent was observed, indicating biodegradation of dye compounds. The fungal isolate effectively utilized the pollutants present in the wastewater and showed good potential for textile effluent bioremediation.

4.4 Seed Germination Response

Green gram (*Vigna radiata*) seeds were treated with different concentrations (20%, 40%, 60%, 80%, and 100%) of fungal-treated textile effluent. Better seed germination, root length, and shoot length were observed at lower concentrations, whereas higher concentrations slightly reduced seed growth. This indicates that fungal treatment decreased the toxicity of the textile effluent and improved its suitability for agricultural application.



Figure 4.3 Seed germination of first day

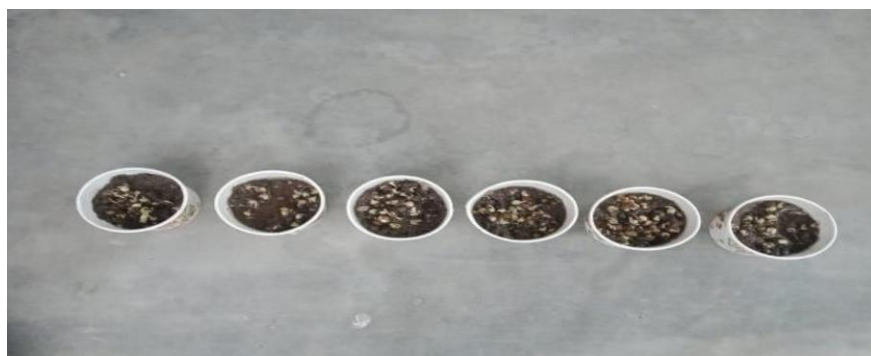


Figure 4.4 Seed germination of second day



Figure 4.5 Seed germination of third day



Figure 4.6 Seed germination of fourth day



Figure 4.7 Seed germination of five days



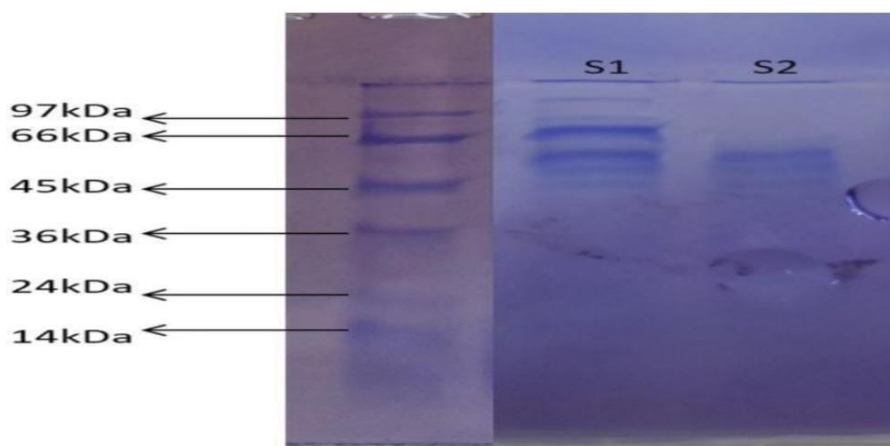
Figure 4.8 Seed germination of six days



Figure 4.9 Seed germination of seven day

4.5 SDS-PAGE Procedure

SDS-PAGE analysis was carried out to study the protein profile of the isolated *Aspergillus species*. Distinct protein bands were observed after staining with Coomassie Brilliant Blue R-250, indicating the production of extracellular proteins involved in the biodegradation process. The protein profile supported the biodegradation potetial to isolate.



5 CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The textile industry is one of the major contributors to environmental pollution due to the discharge of untreated wastewater containing dyes, heavy metals, and other toxic chemicals. These pollutants adversely affect water quality, soil fertility, aquatic organisms, and human health. Therefore, the development of sustainable and eco-friendly wastewater treatment methods is essential to reduce the environmental impact of textile effluents.

Fungi have emerged as promising biological agents for the bioremediation of textile wastewater because of their ability to degrade complex pollutants through enzymatic processes. In addition to pollutant removal, fungal treatment can improve seed germination and promote plant growth, offering both environmental and agricultural benefits. Thus, fungal bioremediation represents an effective, economical, and environmentally sustainable approach for textile effluent management.

5.2 Future scope

- ❖ To isolate fungi from textile effluent samples.
- ❖ To evaluate the ability of fungi to degrade textile pollutants.
- ❖ To study the effect of fungal-treated effluent on seed germination and plant growth.
- ❖ To promote an eco-friendly and cost-effective method for textile wastewater treatment.
- ❖ To contribute towards sustainable environmental protection and agricultural development.

6 CO / PO ATTAINMENT TEMPLATE

This section is included to match the institutional style of the reference thesis. It may be edited as per the current department format and course code requirements.

Table 6.1 CO-PO mapping template for the mini project

Course Outcome (CO)	Description	Mapped PO
CO1	Explain the characteristics of textile effluent, its environmental pollution, and the principles of fungal bioremediation.	PO1, PO2, PO7
CO2	Isolate and identify fungi from textile effluent using standard microbiological techniques.	PO3, PO4, PO5
CO3	Analyze the bioremediation potential of fungal isolates and evaluate seed germination and plant growth promotion. PO4, PO5, PO10	PO4, PO5, PO10
CO4	Interpret the experimental results with awareness of environmental sustainability, ethics, and waste management. PO6, PO7, PO8	PO6, PO7, PO8
CO5	Prepare a structured technical report and present the project findings effectively.	PO9, PO10, PO12

7 REFERENCES

1. Ramalingam S, Saraswathy N, Shanmugapriya S, Shakthipriyadarshani S, Sadasivam S, Sanmugaprakash M. (2010). Decolorization of textile dyes by *Aspergillus tamari*, mixed fungal culture and *Penicillium purpurogenum*. Journal of Scientific and Industrial Research, Vol. 69, pp: 151-153.
2. Hassan, M. A., & El Nemr, A. (2017). Health and Environmental Impacts of Dyes. American Journal of Environmental Science and Engineering, Vol. 1(3), pp: 64-67.
3. Haley AE, Ananthashankar R, Alhattab M, Ramakrishnan VV (2014). Production, Characterization and Treatment of Textile Effluents. Journal of Chemical engineering and process technology, Vol. 5, pp: 1-18.
4. Daneshvar, N., Khataee, A.R., Djafarzadeh, N. (2006). The use of artificial neural networks (ANN) for modeling of decolorization of textile dye solution containing C. I. Basic Yellow 28 by electrocoagulation process, Journal of Hazardous Materials, Vol. 129(2), pp: 116-122.
5. Mazaheri Tehrani, M., Assadi, M. M., & Rashedi, H. (2014). Biodecolorization of Textile Effluents by *Autochthonous* Fungi. Journal of Applied Biotechnology Reports, Vol. 1(4), pp: 161- 165.
6. Gnanadoss, J. J.,(2013). Bioremediation of Textile Dye Using White Rot Fungi, International Journal of Current Research Review, Vol. 5 (3) , pp: 4-11.
7. Salem, S.S.; Fouda,(2021). A Green Synthesis of Metallic Nanoparticles and Their Prospective Biotechnological Applications, Journal of Biological Trace Elements of Research, Vol. 199, pp: 344–370.
8. Hai, F. Ibney., Yamamoto, K., Nakajima, F. & Fukushi, K. (2009). Factors governing performance of continuous fungal reactor during non-sterile operation – Journal of a membrane bioreactor treating textile wastewater . Chemosphere, Vol. 74 (6), pp: 810-817.
9. Bla'nquez, P., Sarra`, M., & Vicent, T. (2008). Development of a continuous process to adapt the textile wastewater treatment by fungi to industrial conditions. Journal of Bioresource Technology, Vol. 99(17), pp: 8002-8009.

10. Martins, M.A.M., Lima, N., Silvestre, A.J.D., Queiroz, M.J. (2003). Comparative studies of fungal degradation of single or mixed bioaccessible reactive azo dyes. *Journal of Biotechnology*, Vol. 89, pp: 91-98.
11. Kaushik, P., Garg, V.K., Singh, B. (2005). Effect of textile effluents on growth performance of wheat. *Journal of Bioresource Technology*, Vol. 96, pp: 1189-1193.
12. Sarker, M. R. H., Razzaque, A., Hoque, M. M., Roy, S., & Hossain, M. K. (2015). Investigation of Effluent Quality from an Effluent Treatment Plant of a Textile Industry, Fakir Knitwear Ltd. Narayanganj, Bangladesh. *Journal of Environmental Science and Natural Resources*, Vol. 8(2), pp: 25-31.
13. Afzal, M., Shabir, G., Tahseen, R., Islam, E., Iqbal, S., Khan, Q. M., & Khalid, Z. M. (2014). Endophytic Burkholderia species. Strain improves plant growth and phytoremediation of soil irrigated with textile effluent. *Journal of environmental science*, Vol. 42(9), pp: 1304–1310.
14. Saroj, S., Dubey, S., Agarwal, P., Prasad, R., Singh, R. P. (2015). Evaluation of the efficacy of a fungal consortium for degradation of azo dyes and simulated textile dye effluents, *Journal of chemical engineering and process technology*, Vol.1, pp:233-243.
15. Mahmood, A., Arshad, M., Khalid, A., & Mahmood, T. (2011). Isolation and screening of azo dye decolorizing bacterial isolates from dye-contaminated textile wastewater. *Journal of Science and Technology*, Vol. 63(4), pp: 723-731.
16. Tony, B. D., Goyal, D., & Khanna, S. (2009). Decolorization of textile azo dyes by aerobic bacterial consortium. *International journal of Biodeterioration and Biodegradation*, Vol. 63, pp: 462–469.
17. Wolff, E., . (2014). Utilization of water treatment plant sludge in structural ceramics. *Journal of Cleaner Production*, Vol.2, pp:228
18. Singh (2017).Biodegradation of synthetic Dyes. A Mycoremediation Approach for Degradation/Decolourization of Textile Dyes and Effluents, *Journal of chemical engineering* ,Vol.3(5), pp: 430-435.
19. Parshetti, G. K., Kalme, S. D., Saratale, R. G., Govindwar, S. P. (2007). Biodegradation of malachite green by textile effluent, *Journal of biochemical engineering*, Vol.54(3), pp: 492-498.
20. Sivakumar, D.,(2014).Effectiveness of Fungi Species in Color Removal from Textile Industry Wastewater. *International Journal of Chemical Technology Research*, Vol. 6(9), 2014, pp: 4366- 4372.

21. Krishnamoorthy R.(2017). Decolourisation and degradation of azo dyes by mixed fungal culture consisted of *Dichotomomyces cejpui* MRCH 1-2 and *Phoma tropica* MRCH 1-3. *Journal of Environmental Chemical Engineering*, vol.6(1) , 2017, pp. 588-595.
22. Gosavi, V. D., & Sharma, S. (2014). *An Journal of Environmental Science, Computer Science and Engineering & Technology*, Vol. 3(1),pp. 29-39.
- 23.Safarzadeh, M. S.,(2001). Advanced oxidation of hazardous and refractory pollutants in wastewaters using the H₂O₂/UV process, *journal of environmental science*,Vol.42(3),pp: 251-325.
- 24.Isık, M., and Sponza, D. T. (2008). Anaerobic/aerobic treatment of a simulated textile wastewater. *Journal of Separation and Purification Technology*, Vol. 60(1),pp: 64–72.
25. Tegli, S., et al. "Decolorization of textile dye effluents by plant-associated fungi. *Journal of Basic Microbiology*, Vol. 54(1), 2014, pp: 57-65.
26. Faryal, R., & Hameed, A. (2005). Isolation and characterization of various fungal strains from textile effluent for their use in bioremediation. *Journal of Botany*, Vol. 37(4), pp: 1003-1008.
- 27.Nilsson, I., Moller, A., Mattiasson, B., Rubindamayugi, M.S.T., and Welander, U. (2006). Decolorization of synthetic and real textile wastewater by the use of *white-rot* fungi. *Journal of Enzyme and Microbial Technology*, Vol. 38, pp: 94-100.
28. Alinsafi A, Evenou F, Abdulkarim EM, Pons MN, Zahraa O, Benhammou A, Yaacoubi A, Nejmeddine A. (2007). Treatment of textile industry wastewater by supported photocatalysis . *Journal of waste water Management*, Vol.74(2), pp: 439-445.
- 29.Kishor, R., Purchase, D., Saratale, G.D.,(2021). Environment friendly degradation and detoxification of Congo red dye and textile industry wastewater by a newly isolated *Bacillus cohnii* (RKS9). *Journal of Environmental Technology and Innovation*, Vol. 2, pp: 101-425.
30. Thoa, L. T. K., Thao, T. T. P., Nguyen-Thi, M.-L., Chung, N. D., Ooi, C. W., Park, S.-M., Lan, T. T., Quang, H. T., Khoo, K. S., Show, P. L., and Huy, N. D.(2023). Microbial biodegradation of recalcitrant synthetic dyes from textile-enriched wastewater by *Fusarium oxysporum* Chemosphere, *Journal of waste water*,Vol.325, pp: 138-392.
31. Rehman, A., Bhatti, H. N., & Athar, H.-u.-R. (2008). Textile Effluents Affected Seed Germination and Early Growth of Some Winter Vegetable Crops. *Water, Air, and Soil Pollution*, *Journal of environmental science*, Vol. 198, pp: 155-163.

32. Shehzadi, M., Fatima, K., Imran, A., Mirza, M. S., Khan, Q. M., and Afzal, M. (2016). Ecology of bacterial endophytes associated with wetland plants growing in textile effluent for pollutant-degradation and plant growth-promotion potentials. *Plant Biosystems - An International Journal Dealing with all Aspects of Plant Biology*, Vol.150(6), pp: 1261-1270.
33. Wawrzekiewicz, M., (2015). Adsorptive removal of acid, reactive, and direct dyes from aqueous solutions and wastewater using mixed silica–alumina oxide. *Journal of Colloid and Interface Science*, Vol. 96, pp: 355–364.
34. McMullan, G., Meehan, C., Conneely, A., Kirby, N., Robinson, T., Nigam, P., Banat, I. M., Marchant, R., and Smyth, W. F. (2001). Microbial decolourisation and degradation of textile dyes. *Journal of Applied Microbiology and Biotechnology*, Vol.56(1),pp: 81-87.
35. Andleeb, S., Atiq, N., Robson, G.D., & Ahmed, S. (2011). An investigation of anthraquinone dye biodegradation by immobilized *Aspergillus flavus* in a fluidized bed bioreactor. *Journal of Environmental Science and Pollution Research*, Vol. 150, pp: 1728–1737.
36. Awomeso, J.A., Taiwo, A.M., Gbadebo, A.M., and Adenowo, J.A. (2010). Studies on the Pollution of Waterbody by Textile Industry Effluents in Lagos, Nigeria. *Journal of Applied Sciences in Environmental Sanitation*, Vol.5(4), pp: 353-359.
37. Espigares M, Crovetto G, Ga'lvez R. (1998). In vitro evaluation of the toxicity of several dithiocarbamates using an *Escherichia coli* growth inhibition bioassay and the acetylcholinesterase inhibition test. *Journal of Environmental Toxicological Water Quality*, Vol.13, pp : 165–174.
38. Mohan, (2017). A case study on textile industrial process, characterization and impacts of textile. *Indian Journal of Scientific Research*, Vol.17(1), pp: 080-084.
39. Khan, M. S., Ahmed, S., Evans, A. E. V., and Chadwick, M. (2009). Methodology for Performance Analysis of Textile Effluent Treatment Plants in Bangladesh .*Journal of Chemical Engineering Research Bulletin*, Vol. 13, pp: 61-66.
40. Balasubramanian, J., Sabumon, P.C., Lazar, J.U., and Ilangovan, R. (2006). Reuse of textile effluent treatment plant sludge in building materials.*Journal of Waste Management*, Vol. 26, pp: 22-28.
41. Kariyajjanavar P, Narayana J, Nayaka YA (2011). Degradation of Textile Wastewater by Electrochemical Method, *Journal of waste water degradation*, Vol. (2), pp: 110.-121.
42. Chatzisyneon, E., Xekoukoulotakis, N. P., Coz, A., Kalogerakis, N., & Mantzavinos, D. (2006). Electrochemical treatment of textile dyes and dyehouse effluents. *Journal of Hazardous Materials*, Vol. 137,pp: 998-1007.

43. Chockalingam N, Banerjee S, Muruhan S.(2019). Characterization of Physicochemical Parameters of Textile Effluents and Its Impacts on Environment. *Journal of Environment and Natural Resources* , Vol. 17(2), pp: 41-53.
44. Yusuff, R.O., & Sonibare, J.A. (2004). Characterization of textile industries effluents in Kaduna, and pollution implications. *International Journal of science and environmental resource*, Vol.6(3), pp: 212-221.
45. Zhong et al. (2011), Preparation, characterization of modified wheat residue and its utilization for the anionic dye removal Desalination, *Journal of plant biology and science*, Vol. 267(2), pp: 193–200.
46. Adeyeye, E. I., & Ayejuyo, O. O. (2002). Assessment of the physicochemical status of a textile industry's effluent and its environment. *Journal of Scientific and Industrial Research*, Vol.45(1), pp: 10-16.
47. Solomon S, Yadessa C, Girma T, Daniel F (2015). Heavy Metal Concentrations and Physicochemical Characteristics of Effluent along the Discharge Route from Hawassa Textile Factory, Ethiopia. *Journal Environmental and Toxicology*, Vol. 5(4), pp: 285-293.
48. Chockalingam N, Banerjee S, Muruhan S.(2019). Characterization of Physicochemical Parameters of Textile Effluents and Its Impacts on Environment. *Journal of Environment and Natural Resources* , Vol.17(2), pp: 41-53.
49. Madhav, S., Ahamad, A., Singh, P., & Mishra, P. K. (2018). Textile industry. Wet processing, environmental impacts, and effluent treatment methods. *Journal of Environmental Quality Management*, Vol.27, pp: 31-41.
50. Nath, (2007). Combinatorial effects of distillery and sugar factory effluents in crop plants. *Journal of Environmental Biology*, Vol. 28, pp: 577-582.